



Java

*Hibernate e
Sqlite*

PAULO R T CÂNDIDO

Criar pasta e abri-la no VSC

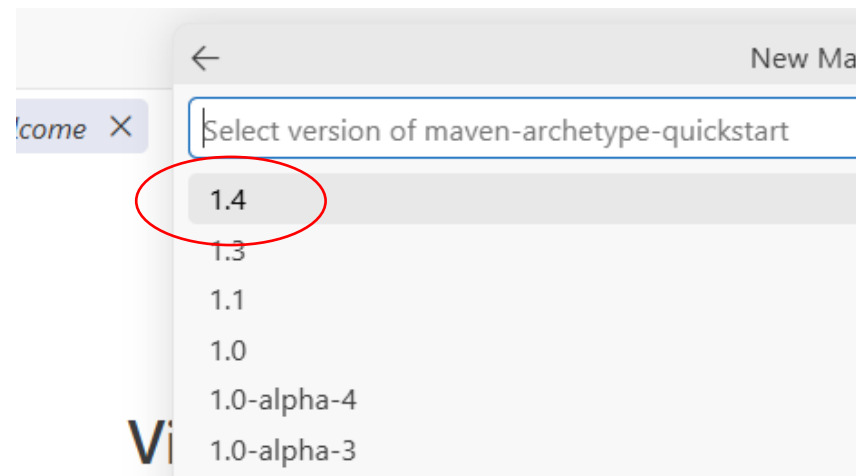
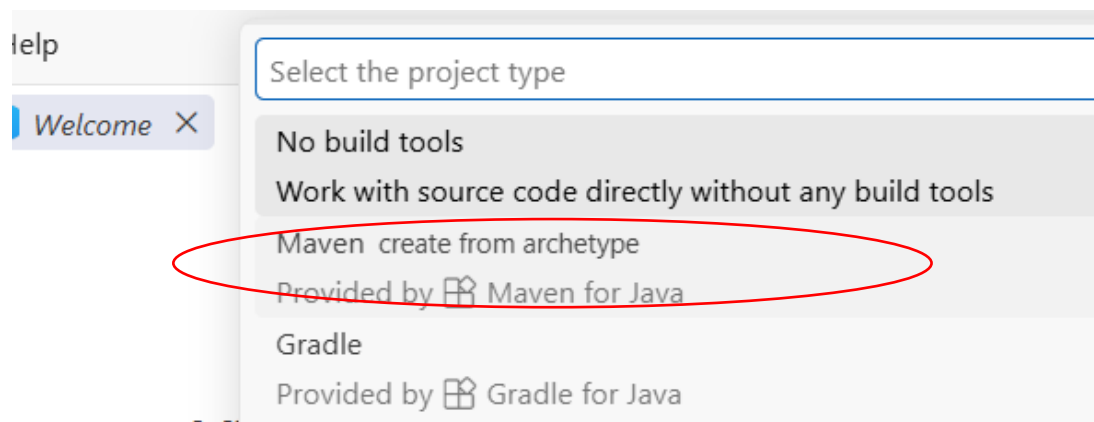
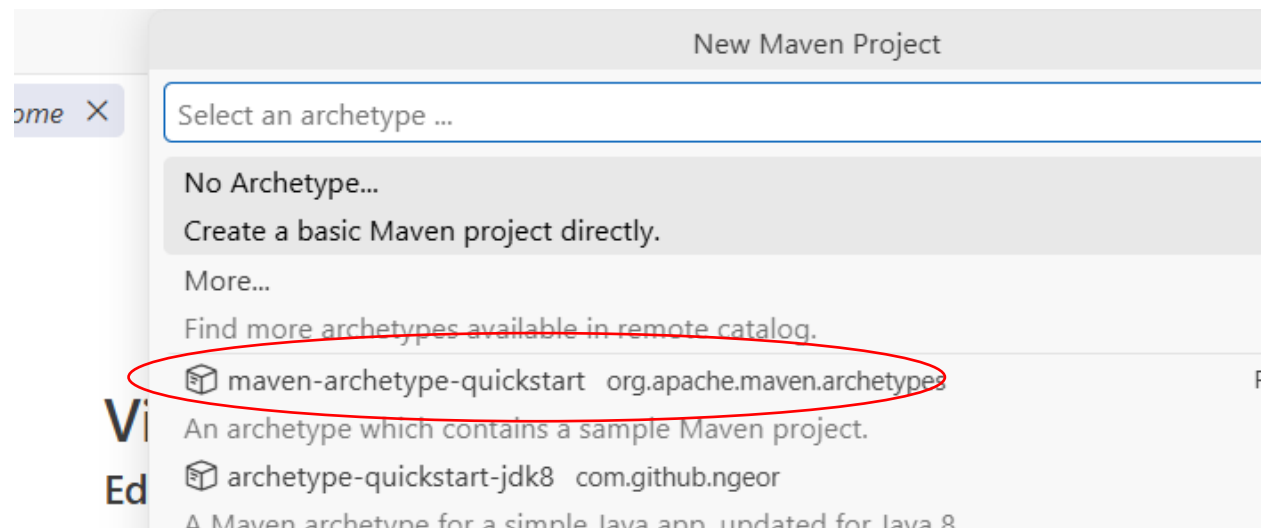
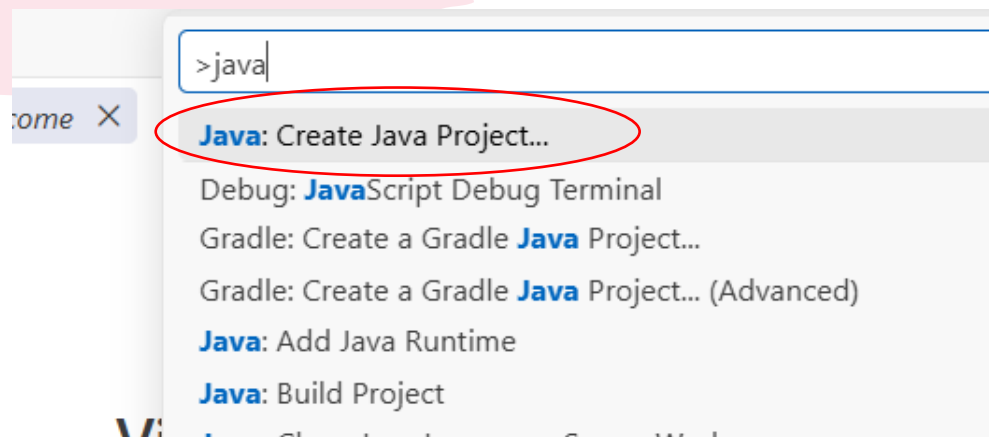
c:\javaproj\poojpa

Explorer

▼ POOJPA

Criar Projeto com Maven

No VSC pressionar CTRL + SHIFT + P



Digite ENTER para as próximas perguntas e selecione a pasta aberta no VSC

Editar arquivo pom.xml – Versão Java



The screenshot shows the Visual Studio Code interface with a project named POOJPA. The file explorer on the left shows the project structure, including a demo folder with a pom.xml file. The editor displays the content of pom.xml, which is a Maven project configuration. The code is as follows:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org
4   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org
5   <modelVersion>4.0.0</modelVersion>
6
7   <groupId>com.example</groupId>
8   <artifactId>demo</artifactId>
9   <version>1.0-SNAPSHOT</version>
10
11   <name>demo</name>
12   <!-- FIXME change it to the project's website -->
13   <url>http://www.example.com</url>
14
15   <properties>
16     <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
17     <maven.compiler.source>26</maven.compiler.source>
18     <maven.compiler.target>26</maven.compiler.target>
19   </properties>
--
```

A red circle highlights the value '26' in the `<maven.compiler.target>26</maven.compiler.target>` property, indicating the Java version used for compilation.

Editar arquivo pom.xml – Hibernate e Sqlite

```
21 <dependencies>
22   <dependency>
23     <groupId>junit</groupId>
24     <artifactId>junit</artifactId>
25     <version>4.11</version>
26     <scope>test</scope>
27   </dependency>
28
29   <!-- 1. Hibernate Core (ORM + JPA) -->
30   <dependency>
31     <groupId>org.hibernate.orm</groupId>
32     <artifactId>hibernate-core</artifactId>
33     <version>6.6.5.Final</version>
34   </dependency>
35
36   <!-- 2. Driver JDBC do SQLite -->
37   <dependency>
38     <groupId>org.xerial</groupId>
39     <artifactId>sqlite-jdbc</artifactId>
40     <version>3.45.1.0</version>
41   </dependency>
42
43   <!-- 3. Dialeto do SQLite para o Hibernate 6+ -->
44   <dependency>
45     <groupId>org.hibernate.orm</groupId>
46     <artifactId>hibernate-community-dialects</artifactId>
47     <version>6.6.5.Final</version>
48   </dependency>
49
50 </dependencies>
```

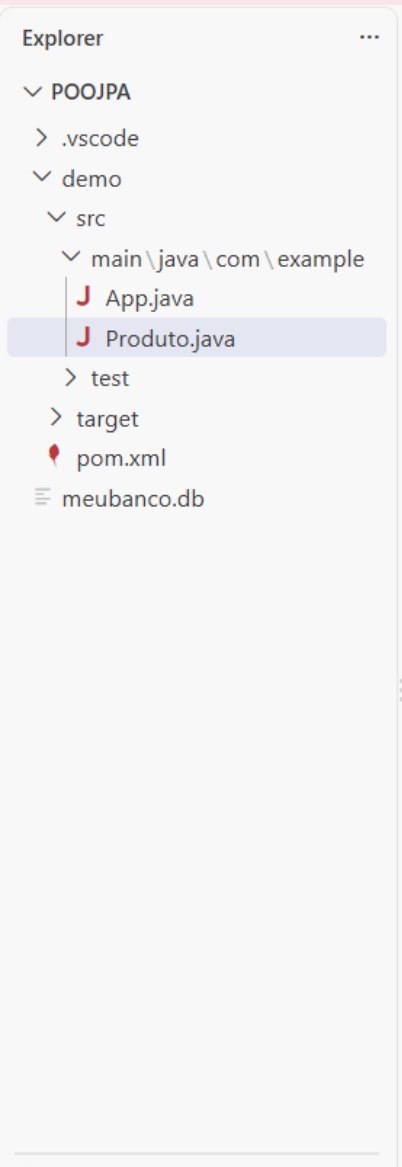
 A build file was modified. Do you want to synchronize the Java classpath/configuration?  

Source: Language Support for Java(TM) by Red ..

Yes Always Never

Quando salvar!!!

Definir entidade mapeada

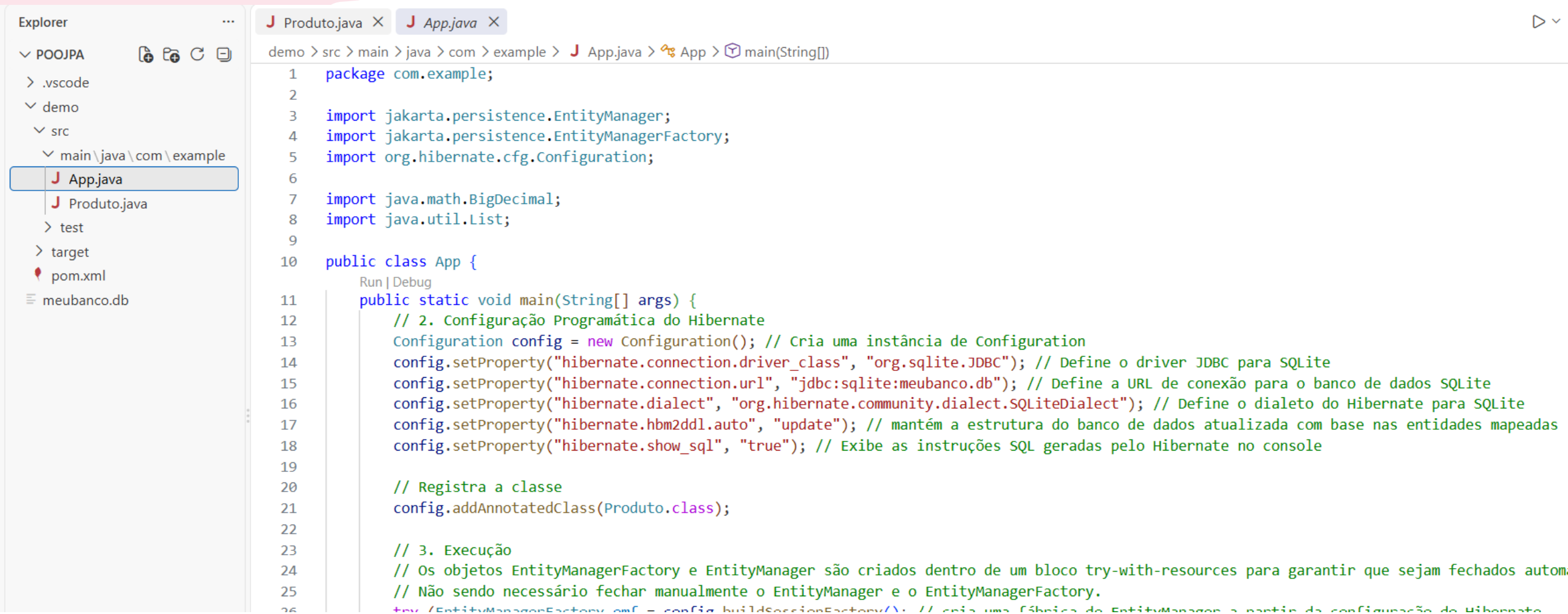


J Produto.java X

demo > src > main > java > com > example > J Produto.java > Produto > id

```
1 package com.example;
2
3 import java.math.BigDecimal;
4
5 import jakarta.persistence.Entity;
6 import jakarta.persistence.GeneratedValue;
7 import jakarta.persistence.GenerationType;
8 import jakarta.persistence.Id;
9
10 @Entity // Indica que esta classe é uma entidade JPA e será mapeada para uma tabela no banco de dados
11 public class Produto {
12     @Id // Indica que o campo 'id' é a chave primária da entidade
13     @GeneratedValue(strategy = GenerationType.IDENTITY) // Especifica que o valor do campo 'id' será gerado automaticamente pelo banco de dados,
14                                                         // utilizando a estratégia de incremento automático
15     private Long id;
16     private String nome;
17     private BigDecimal preco;
18
19     public Produto() {}
20
21     public Produto(String nome, BigDecimal preco) {
22         this.nome = nome;
23         this.preco = preco;
24     }
25
26     @Override
27     public String toString() {
28         return "Produto{id=" + id + ", nome='" + nome + "', preco=" + preco + "}";
29     }
30 }
31
```

Exemplo inserção e consulta



The image shows a VS Code editor interface. On the left is the Explorer sidebar showing a project structure: POOJPA, demo, src, and main\java\com\example. The file App.java is selected. The main editor area shows the code for App.java, which includes package declarations, imports for Jakarta Persistence and Hibernate, and a main method that configures Hibernate for SQLite and registers the Produto class.

```
1 package com.example;
2
3 import jakarta.persistence.EntityManager;
4 import jakarta.persistence.EntityManagerFactory;
5 import org.hibernate.cfg.Configuration;
6
7 import java.math.BigDecimal;
8 import java.util.List;
9
10 public class App {
11     Run | Debug
12     public static void main(String[] args) {
13         // 2. Configuração Programática do Hibernate
14         Configuration config = new Configuration(); // Cria uma instância de Configuration
15         config.setProperty("hibernate.connection.driver_class", "org.sqlite.JDBC"); // Define o driver JDBC para SQLite
16         config.setProperty("hibernate.connection.url", "jdbc:sqlite:meubanco.db"); // Define a URL de conexão para o banco de dados SQLite
17         config.setProperty("hibernate.dialect", "org.hibernate.community.dialect.SQLiteDialect"); // Define o dialeto do Hibernate para SQLite
18         config.setProperty("hibernate.hbm2ddl.auto", "update"); // mantém a estrutura do banco de dados atualizada com base nas entidades mapeadas
19         config.setProperty("hibernate.show_sql", "true"); // Exibe as instruções SQL geradas pelo Hibernate no console
20
21         // Registra a classe
22         config.addAnnotatedClass(Produto.class);
23
24         // 3. Execução
25         // Os objetos EntityManagerFactory e EntityManager são criados dentro de um bloco try-with-resources para garantir que sejam fechados automa
26         // Não sendo necessário fechar manualmente o EntityManager e o EntityManagerFactory.
27         try (EntityManagerFactory emf = config.buildSessionFactory(); // cria uma fábrica de EntityManager a partir da configuração do Hibernate
```

Projeto completo em poojpa.zip

Instalar Sqlite Viewer

